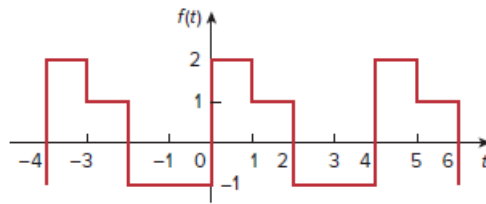


Problem

Find the exponential Fourier series for the function in Fig. 17.81.

Figure 17.81:



Step-by-step solution

Step 1 of 3

Refer to the Figure 17.81 in the text book.

Write the expression of signal $f(t)$.

$$f(t) = \begin{cases} 2 & 0 < t < 1 \\ 1 & 1 < t < 2 \\ -1 & 2 < t < 4 \end{cases}$$

The time period of signal is, $T = 4$

Determine the fundamental frequency.

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{4} \\ &= \frac{\pi}{2} \end{aligned}$$

Step 2 of 3

Determine the exponential Fourier coefficient.

$$\begin{aligned}
 C_n &= \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt \\
 &= \frac{1}{4} \left[\int_0^1 (2) e^{-jn\omega_0 t} dt + \int_1^2 (1) e^{-jn\omega_0 t} dt + \int_2^4 (-1) e^{-jn\omega_0 t} dt \right] \\
 &= \frac{1}{4} \left[2 \int_0^1 e^{-jn\omega_0 t} dt + \int_1^2 e^{-jn\omega_0 t} dt - \int_2^4 e^{-jn\omega_0 t} dt \right] \\
 &= \frac{1}{4} \left[2 \left[\frac{e^{-jn\omega_0 t}}{-jn\omega_0} \right]_0^1 + \left[\frac{e^{-jn\omega_0 t}}{-jn\omega_0} \right]_1^2 - \left[\frac{e^{-jn\omega_0 t}}{-jn\omega_0} \right]_2^4 \right] \\
 &= -\frac{1}{4jn\omega_0} \left[2[e^{-jn\omega_0} - e^0] + [e^{-j2n\omega_0} - e^{-jn\omega_0}] - [e^{-j4n\omega_0} - e^{-j2n\omega_0}] \right] \\
 &= -\frac{1}{4jn\omega_0} \left[2e^{-jn\omega_0} - 2 + e^{-j2n\omega_0} - e^{-jn\omega_0} - e^{-j4n\omega_0} + e^{-j2n\omega_0} \right] \\
 &= -\frac{1}{4jn\omega_0} \left[e^{-jn\omega_0} + 2e^{-j2n\omega_0} - e^{-j4n\omega_0} - 2 \right]
 \end{aligned}$$

Substitute $\frac{\pi}{2}$ for ω_0 .

$$\begin{aligned}
 C_n &= -\frac{1}{4jn\left(\frac{\pi}{2}\right)} \left[e^{-jn\left(\frac{\pi}{2}\right)} + 2e^{-j2n\left(\frac{\pi}{2}\right)} - e^{-j4n\left(\frac{\pi}{2}\right)} - 2 \right] \\
 &= \frac{j}{2n\pi} \left[e^{-jn\left(\frac{\pi}{2}\right)} + 2e^{-jn\pi} - e^{-j2n\pi} - 2 \right] \\
 &= \frac{j}{2n\pi} \left[e^{-jn\left(\frac{\pi}{2}\right)} + 2e^{-jn\pi} - 1 - 2 \right] \quad \left[\text{Since } e^{-j2n\pi} = 1 \right] \\
 &= \frac{j}{2n\pi} \left[e^{-jn\left(\frac{\pi}{2}\right)} + 2e^{-jn\pi} - 3 \right]
 \end{aligned}$$

Step 3 of 3

Write the Fourier series for the signal $f(t)$.

$$\begin{aligned}
 f(t) &= \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \\
 &= \sum_{n=-\infty}^{\infty} \left(\frac{j}{2n\pi} \left[e^{-jn\left(\frac{\pi}{2}\right)} + 2e^{-jn\pi} - 3 \right] \right) e^{jn\omega_0 t} \\
 &= \frac{j}{2\pi} \sum_{n=-\infty}^{\infty} \left(\frac{e^{-jn\left(\frac{\pi}{2}\right)} + 2e^{-jn\pi} - 3}{n} \right) e^{jn\omega_0 t}
 \end{aligned}$$

Therefore, the Fourier series of $f(t)$ is

$$f(t) = \frac{j}{2\pi} \sum_{n=-\infty}^{\infty} \left(\frac{e^{-jn\left(\frac{\pi}{2}\right)} + 2e^{-jn\pi} - 3}{n} \right) e^{jn\omega_0 t}$$